

Speech Enhancement Using Residual Convolutional Neural Network

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Abstract—This paper deals with Deep Speech Enhancement which uses Deep Neural Networks to denoise a noisy speech to produce a clean speech thus removing almost all of the background noise. The method proposed here uses two dimensional convolutional neural networks with residual connections which take advantage of two key facts i.e the non linear functions learned by convolution neural network and the linearity introduced due to residual networks. These factors help remove noise even in low SNR conditions where usually most of the other speech enhancement models fail. The model proposed is fast, robust and lightweight and requires less training as compared to other deep learning models employed for the same task. The model is not expected to produce state of the art results but produces reasonable results keeping in mind the resource limitations and the small size of the model. The model is built upon previous research done in the field and main aim is the reduction in model size while simultaneously preserving the performance.

Keywords—Speech Enhancement, Speech Denoising, Convolutional Neural Network, SNR, Noise Reduction, Noise Distortion, STFT, ISTFT

I. INTRODUCTION

Speech enhancement has been long used as preprocessing step in various speech-related applications such as hearing aids [1, 2], speech coding [3], cochlea implants [4, 5] and automatic speech recognition [6]. Throughout the years extensive research has been carried out in this field starting from the basic techniques of Weiner filtering [7], Spectral Subtraction [8], non-negative matrix factorization [9]. These methods fail to learn any useful from the data but provide a blind estimation on predefined speech and noise models. Their model assumptions do not match real world complex non stationary noise, which leads to failure of these techniques in real world applications.

Many deep neural networks have been explored in this domain showing promising result but almost all of them suffer the curse of complexity and resource usage. Traditionally, Fully Connected Deep Neural Networks were explored for the speech enhancement task but the size of neural network usually ends up in hundreds of megabytes which becomes difficult to use in embedded systems.

The Convolutional Neural Networks however on the other hand consist of lesser number of parameters due to their weight sharing property. The models converge faster and are less resource exhaustive and can be embedded in portable devices. It has been seen that, historical models like Fully Connected Deep Neural Networks (DNN) use 20-30 times more parameters for same number of layers with same number of inputs, so convolutional layer based reduce parameters and decrease the training time to get to same results and also CNN

works better in the case of finding local dependencies and leads to produce far better results than that of Dense networks.

The paper attempts to find a memory efficient denoising algorithm that removes background noise from unseen test samples which include unseen speakers and unseen noise conditions. Unseen meaning the samples that were not used during the training of the network to ensure that the model generalizes well on real world conditions and does not stick to the training specific conditions.

The paper consists of two convolutional neural network architectures on which the results are evaluated. The proposed architecture which is a convolutional neural network with skip connections and a similar model without the skip connections. Both these architectures are trained with and without dropouts to evaluate the performance measures and the proposed model without the dropouts outperforms other models based on the evaluation metrics Noise Reduction and Noise Distortion. Paper primarily focuses on these metrics derived from mean-square-error because we use this function to train our neural networks.

Paper is divided into following sections. Section II contains problem description. Section III proposes a model to solve the formulated problem. Section IV contains various experiments that were carried out along with the setups of those experiments. Section V contains the results of the carried out experiments. Finally section VI contains the conclusions drawn.

II. PROBLEM FORMULATION

Let $x \in X^{f \times t}$ and $y \in X^{f \times t}$ be the parallel noisy along with its corresponding clean spectrogram, where t is the length of the spectrogram and f represents the dimensions of frequency bins in the spectrogram. Let given training set $\text{Train} = \{(x_m, y_m)\}_{m=1 \text{ to } n}$ of n pairs of clean and noisy signal spectrograms, the problem can be formalized as finding a mapping between noisy and clean signal i.e. $Z : X^{f \times t} \rightarrow X^{f \times t}$, which maps noisy utterances to clean one, by the way of minimizing L2 norm formally called mean-square-error between input and output signal, to find the best possible weights w for neural model,

$$\Delta w = \frac{1}{n} \sum_{m=1}^n \|Z(x_m) - y_m\|^2$$

Δw is the loss of our model which tells which tells that how far is our model's predictions from the expected outputs.

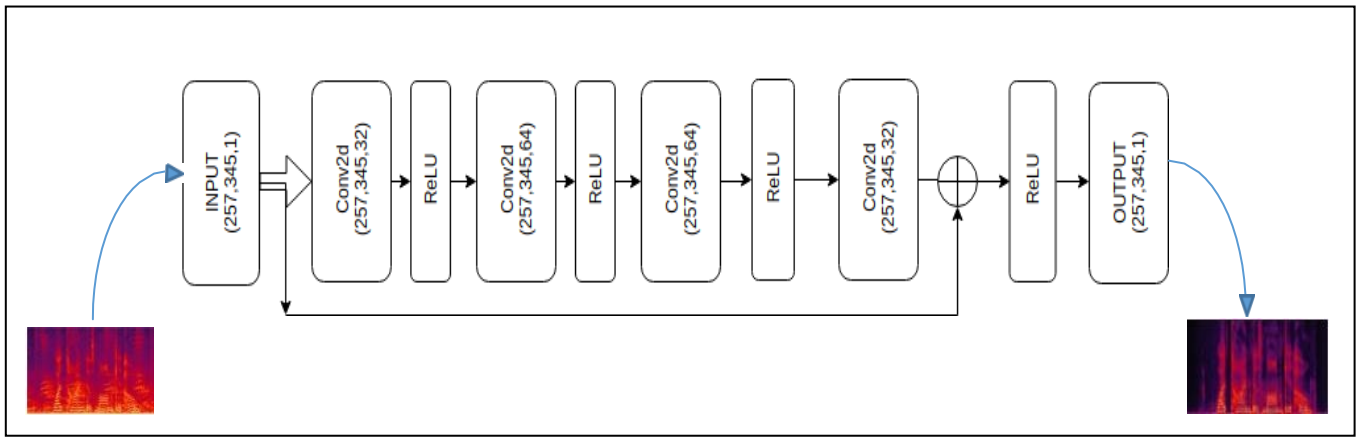


Figure 1 Model Architecture. RCNN consists of 4 convolution layers with same padding and ReLU activation. Noisy speech spectra is given as input and clean speech spectra is obtained at the output.

III. RESIDUAL CONVOLUTIONAL NEURAL NETWORK

Convolutional Neural Networks (CNN) are capable of exploring local patterns in the spectrogram from both, frequency and temporal domains. CNN consists of only one type of layer, convolutional layers. In a case of convolutional layer, short term fourier transformation (stft) is passed as input to layer and each convolutional layer takes small rectangular grid as inputs from the previous layer which are then multiplied with the weights matrix w . The Localized filter or weights matrix, will be copied across entire input space to detect specific types of local patterns. Neurons sharing the same weights will compose a feature map. In our implementation for speech enhancement, the input space is a 2-D plane with frequency and time axis. No pooling layer and same padding is added based on kernel so that network doesn't lose any of its valuable features.

$$Padding = \frac{(Kernel - 1)}{2} \quad Output = \frac{(Width - Kernel + 2 * Padding)}{stride} + 1$$

Here, padding is added based on kernel and output is the shape of the output window produced based on width of previous window, kernel size, stride and padding of the convolutional layer. Network also features skip connections, connecting input and output layers. This is done because the input and output produced share the same structure and many features could be lost while reproducing the output at last year due to vanishing gradient. Skip connections help in preserving the information so that there is not degradation of our speech signal or we preserve good features. The skip connections allow gradients to move deeper into the network thus ultimately improving the results.

$$Y^n = (w^{n-1,n} * a^{n-1} + b^n) + (w^{0,1} * x^0 + b^1) \\ Y^{out} = ReLU(Y^n)$$

Where Y^n is the output of last convolution layer in which $w_{n-1,n}$ is weight of the last layer and a_{n-1} is the activation of its previous layers and b_n is the bias of the last layer which are then added to weights $w^{0,1}$, inputs x^0 and bias b^1 of the first layer. Y_{out} is the output of last layer which is produced by passing Y_n in the Rectified Linear Unit (ReLU) non-linearity.

We propose to do so with a Residual Convolutional Neural Network (RCNN) shown in Fig.1. The neural network architecture used in our model only uses convolutional layers. The model thus only tries to learn the temporally close correlations in the data input. Since only convolutional layers are used this reduces the number of training parameters which would have been considerably more if we had used dense layers and as a result our training time is reduced. The network produces the same dimension outputs as the input and behaves like an autoencoder. In the encoding stage, the input signal is strided across in several convolutional layers to get an expanded form of signal and in decoding stage we again use convolutional layers for getting same output shape as input.

IV. EXPERIMENTATION

A. Dataset

The publicly available Edinburgh dataset [12] for training and testing the model. Amount of data and speakers fit the problem description. The entire dataset is resampled at 16kHz. The Training set contains 28 speakers with 400 sentences per speaker. There are 10 types of noise with 4 SNR conditions {15, 10, 5, 0} dB. Thus there are 10 sentences per speaker in each noise and SNR condition. The test set contains unseen speakers and noise conditions which were not used during the training of the model. It contains 2 speakers and 5 types of noise with 4 SNR conditions each {17.5, 12.5, 7.5, 2.5} dB. Thus there are 20 sentences for each condition per speaker.

B. Evaluation Metrics

The performance of our model is evaluated in terms of Noise Reduction and Speech Distortion [23] as they are the most used estimators for speech quality.

C. Preprocessing

Before any actual training of the model starts we need to extract features from our raw audio wave which will be utilized to train the model. We use 2 s long noisy speech sentences to train our model and 512-point Hamming window with 75% overlap.

The unclean speech is presented as input into the model such that the least amount of features containing maximum possible information about the signal are presented. Switching from time domain to frequency domain does the task. For this, STFT (Short Term Fourier Transform) is employed.

At this point, our signal is represented by a complex FFT array whose magnitude and phase values can be used as input to the neural network. However, so far, we seem to have only increased the size of the input to our neural network.

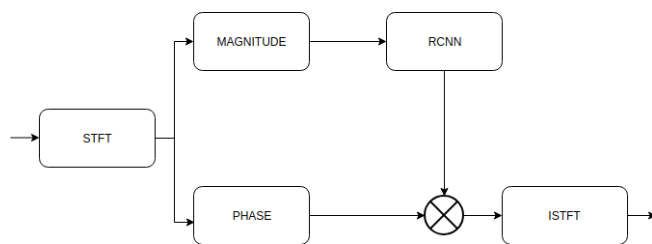


Figure 2 Preprocessing steps applied to raw audio waveform. The waveform is given as input to stft and the predicted results are multiplied with the phase from noisy signal and then the ISTFT is employed in order to produce the final output audio file.

The phase of the signal is ignored. The neural model is optimised only on the magnitude of the audio signal without taking into context the phase which is used later for producing the clean audio file by combining the output from the neural model and phase from the noisy signal. This method however, puts a limit on how good of a results can be obtained following this approach.

D. Training of model

Model is basically trained just for 200 epochs and it gives reasonable results, smaller layered network is also used because of lack of computational requirements as model is trained on google colab. It has been seen that by adding the skip connection in the model, it produces better signal than that of the model without skip connections, because skip connection helps conserving low level features and doesn't let them vanish. While training RCNN, it has been seen that addition of dropout layer in the model leads to deterioration of the quality of signals because it drops out useful features too while dropping random features, which can't be easily recovered and leads to distorted signal and also needs more time to converge to given set of weights as RCNN without dropout. Model is trained on different kinds of non-linearities like elu, selu, prelu and relu, in which relu came out with most pleasing results because it is not prone to problem like vanishing gradients relu substantially reduces

the computational cost for training of the net. This allows the training of larger nets with more parameters at the same computational cost, thus leading to higher capacity, and often higher test set accuracy as well. Different kinds of optimizations like SGD, Adagrad, Adam are applied on learning rate of 0.0005 in which Adam was the winning candidate.

V. RESULTS

The paper explores the potential of convolutional neural networks in speech enhancement application. The experiments are carried out on two base network architectures one with skip connections and one without it. For novelty, we also perform analysis to validate the results of using dropouts while training our models. As listed in table 1 we can see that overall dropouts fail to augment the network's capability of mapping the spectra from noisy to clean efficiently. The proposed network with skip connections trained without dropouts outperforms other architectures and gives reasonable results keeping in mind the hardware and resource limitations present along with the condition for developing a light model that converges faster and uses less data.

TABLE I. NR and SD PERFORMANCE OF MODELS

Model	Noise Reduction	Speech Distortion
CNN with dropout	0.854	0.142
CNN without dropout	0.834	0.055
RCNN with dropout	0.892	0.054
RCNN without dropout	0.969	0.042

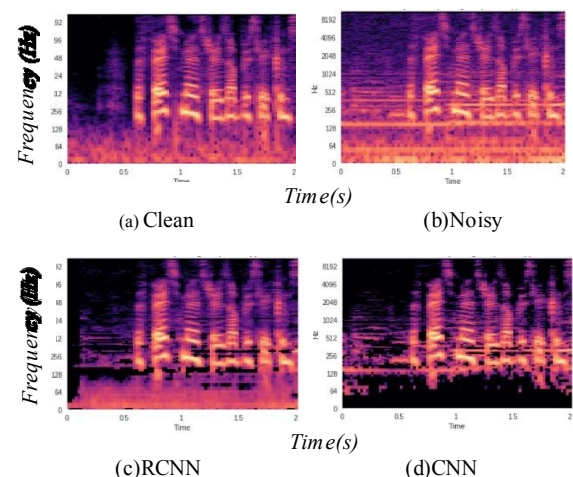


Figure 3 The Spectrogram of (a) Clean speech, (b) Noisy speech (c) enhanced speech by RCNN without dropouts, and (d) enhanced speech by CNN without dropouts

TABLE II. MODEL SIZE COMPARISON

Model	Size (mb)
Deep Neural Network Based Speech Enhancement	100-150
Reinforcement Learning based Speech Enhancement	50-70
Speech Enhancement using Generative Adversarial Network (SEGAN)	40-60
Convolutional AutoEncoder for Speech Enhancement	10-20
Residual Convolutional Neural Network For speech Enhancement	0.6-0.7

The size of our model is considerably smaller than the other proposed networks without compromising much of the

performance levels. This is a huge step forward towards integrating neural network based speech enhancement models in wearable devices.

VI. CONCLUSION

The focus of this paper is speech enhancement that is solved through the use of the neural networks. The work here focuses on a neural network architecture that is easy to implement, train and use in real world thus eliminating the major drawbacks of the neural networks. Proposed model reduces noise by 96.9% in the speech and produces speech distortion of 4.2%. We see 99.4% decrease in model size from the best performing model and only a decrease of 10% in performance. This study serves as a pioneering work for exploiting the full potential of neural networks without focusing on increasing their size for enhanced performance. In the future work, more noise types and SNR levels can be considered to build a more robust network.

VII. ACKNOWLEDGMENT

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